



Meningitis

Prof. Dr. Norma Jung, Uniklinik Köln

ABS Kurs

Conflict of interest (2023 – 2026)

- Lecture fees: AbbVie, Cogitando, GSK, Infectopharm, Medupdate, Pfizer, Phaion, Shionogi, Tillotts
- Travel grants: Pfizer
- Fees for advisory board: MSD, Shionogi
- Honorarium for editorial board: Springer (“Die Innere Medizin”), Infectopharm (“Podcast Consilium Infectorum”)
- Scientific Board: Streamed-up -InfektioLive

Überblick

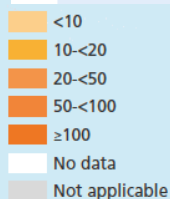
- Globale Daten
- Guidelines
 - Empirische Therapie
- Der besondere Fall
- Choosing wisely initiative– “Klug entscheiden”
 - Meningitis

WHO goal for Meningitis – 2030

Fig. 1. Incidence rates of all-cause meningitis per 100 000 population by country in 2017

Visionary goals by 2030

- ✓ Eliminate bacterial meningitis epidemics⁵
- ✓ Reduce cases of vaccine-preventable bacterial meningitis by 50% and deaths by 70%⁶
- ✓ Reduce disability and improve quality of life after meningitis due to any cause



Data Source: Global Burden of Disease Collaborative Network.
Global Burden of Disease Study 2019. IHME, 2020.
Map Production: WHO GIS Centre for Health, DNA/DDI

Aetiology of Meningitis – globally

GBD 2019 Meningitis and Antimicrobial Resistance
Collaborators*, Lancet Neurol, 2023

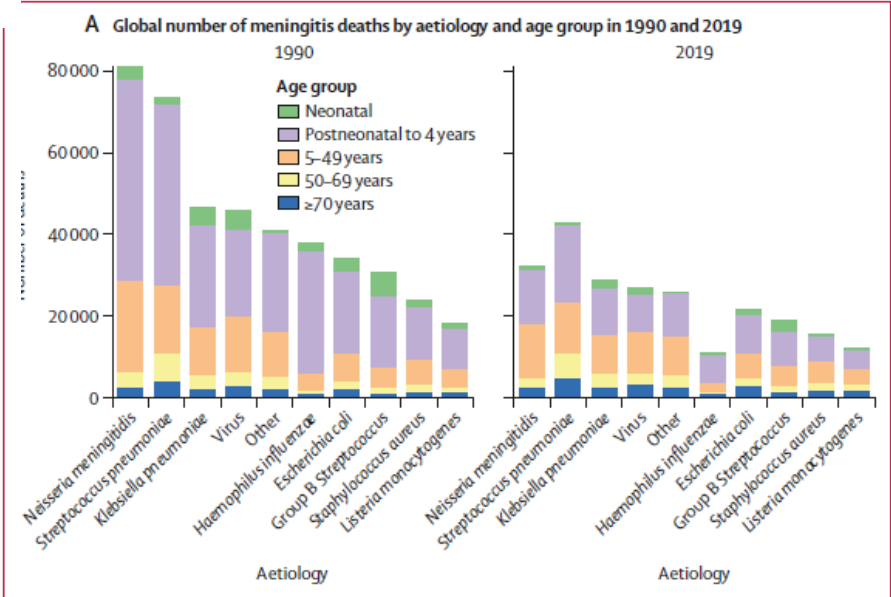
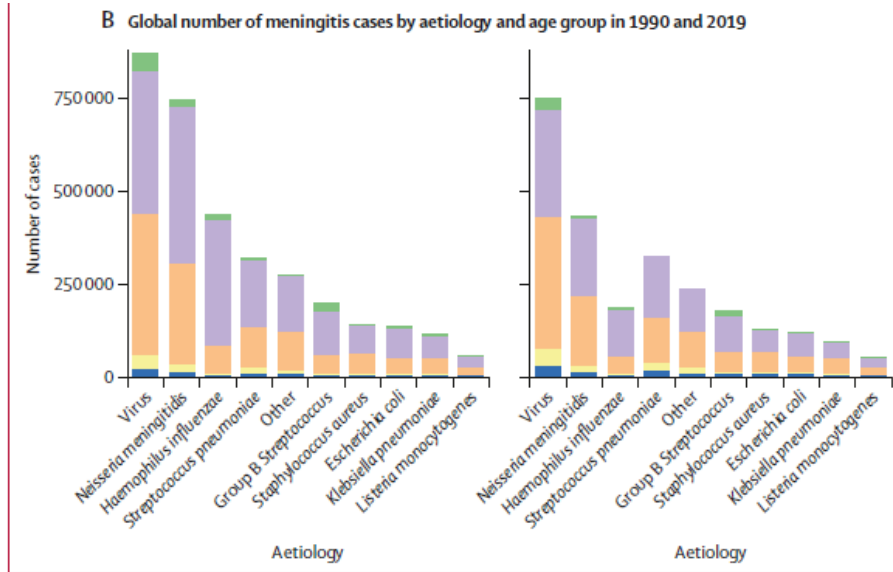


Figure 2: Global number of meningitis deaths (A) and cases (B) by aetiology and age group in 1990 and 2019

- *N. meningitidis*
- *S. pneumoniae*



Guidelines

Tunkel, 2004; Van de Beek 2016, Pfister 2023,

IDSA GUIDELINES

Practice Guidelines for the Management of Bacterial Meningitis

Allan R. Tunkel,¹ Barry J. Hartman,² Sheldon L. Kaplan,³ Bruce A. Kaufman,⁴ Karen L. Roos,⁵ W. Michael Scheld,⁶ and Richard J. Whitley⁷

ESCMID guideline: diagnosis and treatment of acute bacterial meningitis

D. van de Beek¹, C. Cabellos², O. Dzupova³, S. Esposito⁴, M. Klein⁵, A. T. Kloek¹, S. L. Leib⁶, B. Mourvillier⁷, C. Ostergaard⁸, P. Pagliano⁹, H. W. Pfister⁵, R. C. Read¹⁰, O. Resat Sipahi¹¹ and M. C. Brouwer¹, for the ESCMID Study Group for Infections of the Brain (ESGIB)

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Forum der wissenschaftlichen Medizin

Ambulant erworbene bakterielle Meningoenzephalitis im Erwachsenenalter

2004
USA

2016
Europe

2023
Germany



Empirische Therapie der Meningitis

Erreger abhängig vom

- Alter – Neugeborene/Ältere
- Ambulant/nosokomial erworben
- Immun-kompetent/supprimiert

35 jähriger Patient– Meningitis – keine Vorerkrankungen



Welches empirisches Regime wählen Sie?

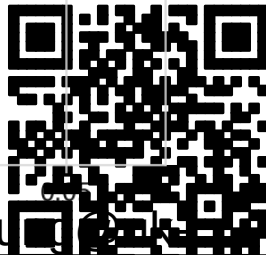
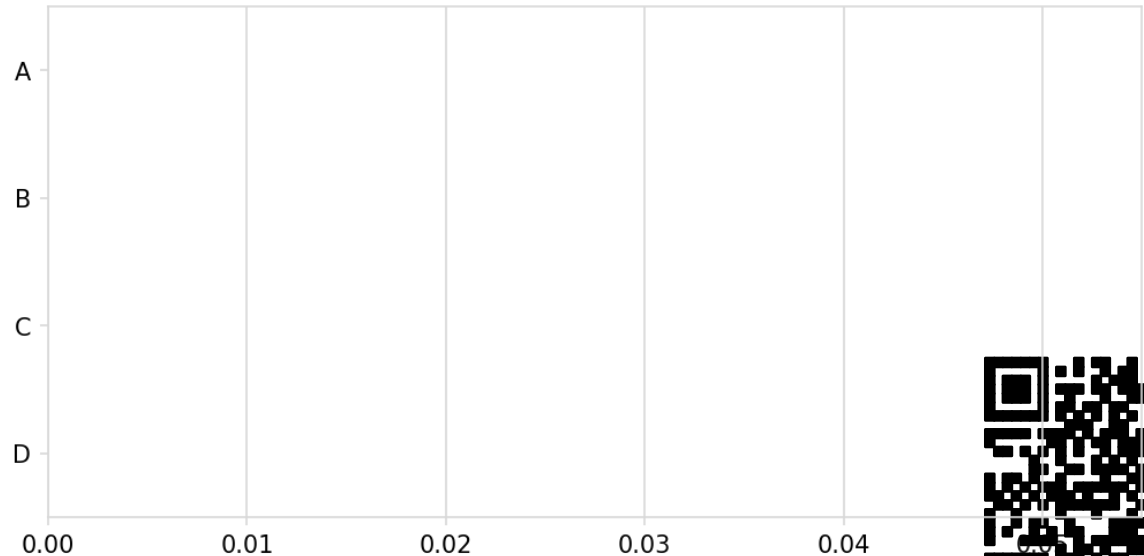
- A) Ceftriaxon
- B) Ampicillin +
Ceftriaxon
- C) Vancomycin +
Ceftriaxon
- D) Vancomycin +
Meropenem

35 jähriger Patient– Meningitis – keine Vorerkrankungen



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Umfrage noch nicht

Guidelines



Practice Guidelines for the Management of Bacterial Meningitis

Allan R. Tunkel,¹ Barry J. Hartman,² Sheldon L. Kaplan,³ Bruce A. Kaufman,⁴ Karen L. Roos,⁵ W. Michael Scheld,⁶ and Richard J. Whitley⁷

Predisposing factor	Common bacterial pathogens	< 1month	Ampicillin + Cefotaxime	<i>L.monocytogenes</i> + <i>S. agalactiae</i> + <i>E. coli</i>
Age				
<1 month	<i>Streptococcus agalactiae</i> , <i>Escherichia coli</i> , <i>Listeria monocytogenes</i> , <i>Klebsiella</i> species			
1–23 months	<i>Streptococcus pneumoniae</i> , <i>Neisseria meningitidis</i> , <i>S. agalactiae</i> , <i>Haemophilus influenzae</i> , <i>E. coli</i>			
2–50 years	<i>N. meningitidis</i> , <i>S. pneumoniae</i>			
>50 years	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>L. monocytogenes</i> , aerobic gram-negative bacilli			
		1 month-50 years	Vancomycin + Cefotaxime	<i>S. pneumoniae</i> <i>N. meningitidis</i>
		>50 years	Vancomycin + Cefotaxime + Ampicillin	+ <i>L. monocytogenes</i>

D. van de Beek¹, C. Cabellos², O. Dzupova³, S. Esposito⁴, M. Klein⁵, A. T. Kloek¹, S. L. Leib⁶, B. Mourvillier⁷, C. Ostergaard⁸, P. Pagliano⁹, H. W. Pfister⁵, R. C. Read¹⁰, O. Resat Sipahi¹¹ and M. C. Brouwer¹, for the ESCMID Study Group for Infections of the Brain (ESGIB)

Guidelines

TABLE 4.1. Empiric antibiotic in-hospital treatment for communi

Patient group	Standard treatment	
	Reduced <i>Streptococcus pneumoniae</i> antimicrobial sensitivity to penicillin	<i>S. pneum</i> susceptib
Neonates <1 month old	Amoxicillin/ampicillin/penicillin plus cefotaxime, or amoxicillin/ampicillin plus an aminoglycoside	
Age 1 month to 18 years	Cefotaxime or ceftriaxone plus vancomycin or rifampicin	Cefotaxim
Age >18 and <50 years	Cefotaxime or ceftriaxone plus vancomycin or rifampicin	Cefotaxim
Age >50 years, or Age >18 and <50 years plus risk factors for <i>Listeria monocytogenes</i> ^a	Cefotaxime or ceftriaxone plus vancomycin or rifampicin plus amoxicillin/ampicillin/penicillin G	Cefotaxim plus am penicillin

^aDiabetes mellitus, use of immunosuppressive drugs, cancer and other conditions causing i

	<i>S. pneumoniae</i> susceptible	<i>S. pneumoniae</i> reduced susceptible	
<1 month	Ampicillin + Cefotaxime	Ampicillin + Cefotaxime	<i>L. monocytogenes</i> + <i>S. agalactiae</i> + <i>E. coli</i>
1 month-50 years	Cefotaxime	Vanco-mycin + Cefotaxime	<i>S. pneumoniae</i> <i>N. meningitidis</i>
> 50 years	Ampicillin+ Cefotaxime	Vanco-mycin + Cefotaxime + Ampicillin	+ <i>L. monocytogenes</i>

Ambulant erworbene bakterielle Meningoenzephalitis im Erwachsenenalter

4.2 Antibiotikatherapie

Tabelle 3: Initiale Antibiotikatherapie der bakteriellen Meningitis (ohne Erregernachweis) bei Erwachsenen

Klinische Konstellation	Empfohlenes Antibiotikaregime
ambulant erworben („community-acquired“), keine Immunsuppression	Cephalosporin Gruppe 3a ¹ plus Ampicillin ²

**Ampicillin +
Cefotaxime**
community acquired

35 jähriger Patient– Meningitis – keine Vorerkrankungen

Diese empirischen Regime sind empfohlen!

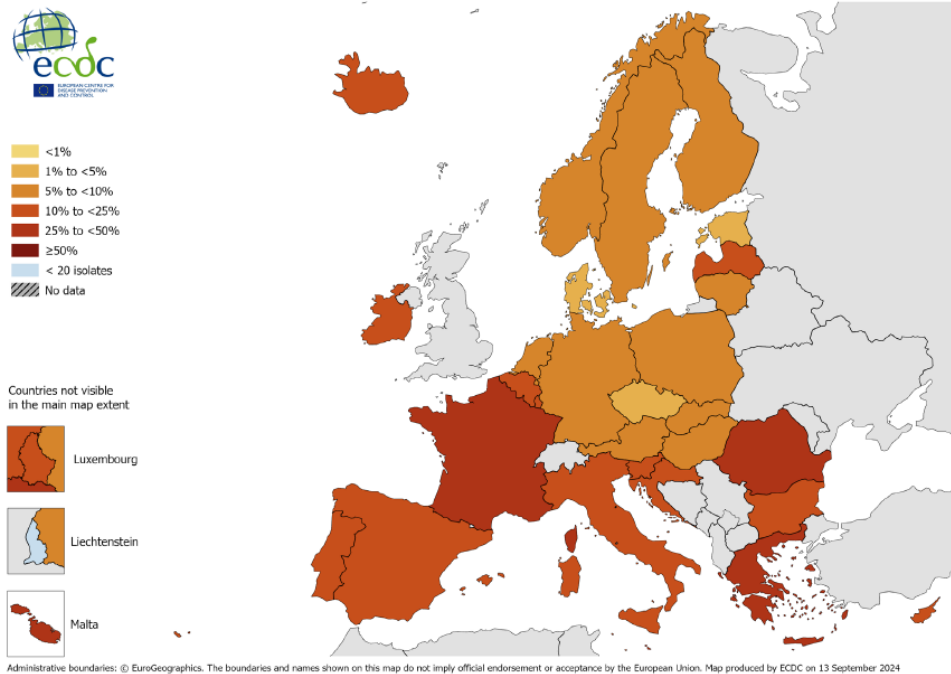
- A) Ceftriaxon
- B) Ampicillin + Ceftriaxon
- C) Vancomycin + Ceftriaxon
- D) Vancomycin + Meropenem

Streptococci pneumoniae- european data – penicillin resistance



ECDC, 2023

Figure 9. *Streptococcus pneumoniae*. Percentage of penicillin^a non-wild type^b invasive isolates, by country, EU/EEA, 2023



Streptococci pneumoniae- deutsche Daten (im Krankenhaus- 2023)



<https://amr.rki.de/Content/Datenbank/ARS/ResistanceOverview.aspx>

	N	S %	I %	R %	95% KI R
Carbapeneme					
Ertapenem	158	100,0	0,0	0,0	0,0 - 2,4
Imipenem	1.747	99,8	0,1	0,1	0,0 - 0,3
Meropenem	2.316	99,8	0,1	0,1	0,0 - 0,3
Cephalosporine					
Cefepim	310	99,0	1,0	0,0	0,0 - 1,2
Cefotaxim	5.332	98,6	1,2	0,3	0,2 - 0,5
Ceftriaxon	4.771	98,3	1,3	0,3	0,2 - 0,5
Cefuroxim	2.947	98,6	0,3	1,1	0,7 - 1,5
Fluorchinolone					
Levofloxacin	5.631	1,5	98,0	0,6	0,4 - 0,8
Moxifloxacin	5.154	99,0	0,8	0,2	0,1 - 0,4
Glykopeptide					
Teicoplanin	2.790	100,0	0,0	0,0	0,0 - 0,1
Vancomycin	5.650	100,0	0,0	0,0	0,0 - 0,1
Makrolide - Lincosamide - Streptogramine					
Azithromycin	1.338	88,1	0,1	11,7	10,1 - 13,6
Clarithromycin	1.279	88,3	0,1	11,6	10,0 - 13,5
Clindamycin	5.623	93,2	0,0	6,8	6,1 - 7,4
Erythromycin	6.059	87,6	0,0	12,3	11,5 - 13,2
Penicilline					
Amoxicillin	1.488	97,4	1,1	1,5	1,0 - 2,2
Amoxicillin/Clavulansäure	2.379	97,8	0,8	1,4	1,0 - 2,0
Ampicillin	4.430	96,3	1,4	2,3	1,9 - 2,8
Ampicillin/Sulbactam	3.042	98,1	0,7	1,2	0,9 - 1,7
Penicillin	6.575	88,9	7,7	3,4	3,0 - 3,9

Sensitivity

- Meropenem (99,8%)
- Ceftriaxon (98,3%)
- Vancomycin (100%)
- Azithromycin (88,1%)
- Ampicillin (96,3%)
- Penicillin (88,9%)

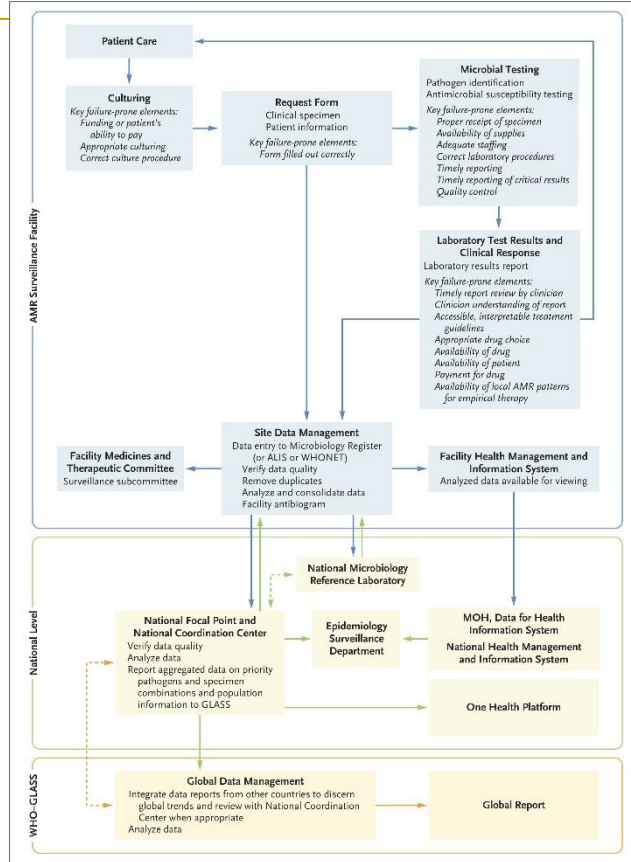
Controlling global - AMR



Goldmann, NEJM, 2024
GLASS

Different levels

- facility
- national
- global



S. pneumoniae



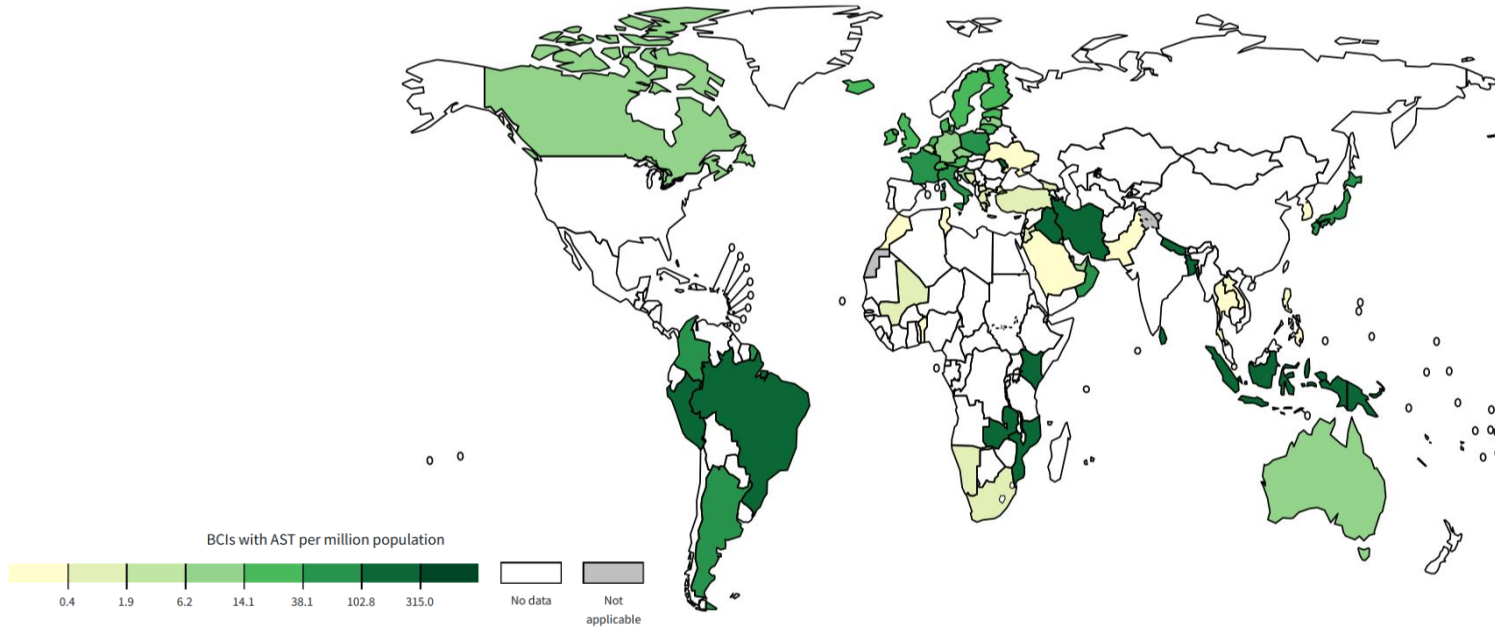
GLASShttps://worldhealthorg.shinyapps.io/glass-dashboard/_w_b2a568d7/#!/amr

2022

Bloodstream

Streptococcus pneumoniae

Penicillins

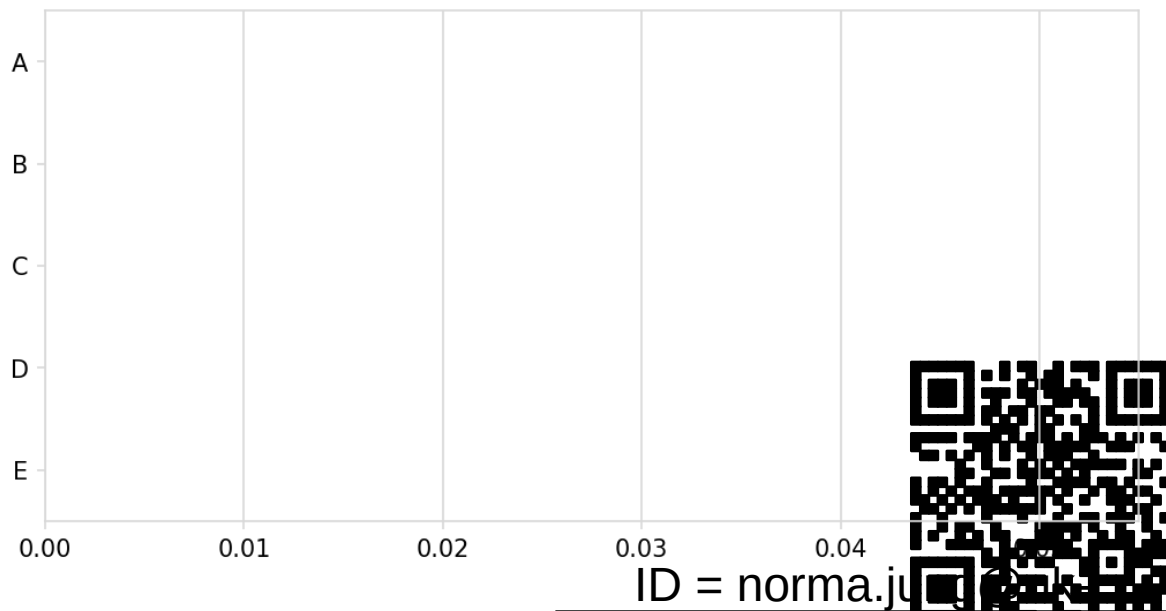


Resistenzniveau – empirische Therapie der Meningitis



Ab welchen % - wählen Sie empirisch eine Alternative?

- A) 5%
- B) 10%
- C) 15%
- D) 20%
- E) 30%



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Umfrage noch nicht

What is the evidence base of used aggregated antibiotic resistance percentages to change empirical antibiotic treatment? A scoping review

Table 2
Overview of criteria used to define thresholds

Source	Threshold	Criteria used ^a					
		Efficacy ^b	Disease severity	Increase in ABR ^c	Adverse drug event	<i>C. difficile</i>	Costs
IDSA guidelines 1999 [2], 2011 [3], EAU guideline 2018 [5], Bader et al. [31]	Cotrimoxazole (UTI) > 20%	Yes	Yes	Yes	Yes	Not mentioned	Not mentioned
IDSA guidelines 2011 [3], EAU guideline 2018 [5], Bader et al. [20]	Ciprofloxacin (UTI) > 10%	Yes	Yes	Yes	Not mentioned	Not mentioned	Not mentioned
Le et al. [16]	Cotrimoxazole (UTI) > 22%	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Yes
Perfetto et al. [18]	Cotrimoxazole (UTI) > 4.3%	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Yes
McKinnell et al. [17]	Cotrimoxazole (UTI) > 10%	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Yes
Sadler et al. [19]	Trimethoprim (UTI) > 30%	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Yes
IDSA and ATS consensus	Macrolides (CAP) > 25%	Not	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned
Bradley et al. [22]	Cephalosporins (meningitis) > 5%	Not mentioned	Yes	Yes	Not mentioned	Yes	Yes
guideline HAP/VAP 2016 [1]		mentioned					Yes
IDSA and ATS consensus guideline VAP 2016 [1]	Gram-negatives (VAP) > 10%	Not mentioned	Yes	Yes	Yes	Yes	Yes
Babela et al. [21]	Macrolides (rhinosinusitis) > 13.8%	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Yes
Kaplan et al. [23]	MRSA (SSTI) > 10–15%	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned
Bradley et al. [22]	Cephalosporins (meningitis) > 5%	Not mentioned	Yes	Not mentioned	Not mentioned	Not mentioned	Not mentioned

What is the evidence base of used aggregated antibiotic resistance percentages to change empirical antibiotic treatment? A scoping review

Threshold recommendations	
1. Antibiotic guideline committees should weigh explicitly the criteria used to come to their antibiotic choice: disease severity, coverage, efficacy, risk of increase in ABR, toxicity, adverse drug effects, and costs) when defining a threshold.	
2. Disease severity of an infection has an important impact on antibiotic choice. The more severe the lower the threshold to switch to an alternative.	
3. Risk of transmission appears to be an important component that requires further thought	
4. Differences in clinical breakpoints used between regions should be considered when defining thresholds until there is global uniformity in the definitions of susceptible and resistant for all bacteria.	
5. Thresholds should be defined as precise as possible and broad ranges (e.g. 10–20%) should be avoided.	
6. When defining a threshold, clear recommendations should be given for one or more alternative antibiotics. The recommendation should also consider de-escalation indications and alternatives for when the antibiogram becomes available.	

Fig. 2. Recommendations to consider when defining thresholds.

Conclusion:

Niedriger Evidenzgrad!

Empfehlung

- systematic multidisciplinary approach
- möglichst präzise (keine Ranges)
- für verschiedene Infektionen
- Schwere der Erkrankung, Effektivität, Resistenzentwicklung, Toxizität, Kosten
- Cave: unterschiedl. Breakpoints

Patientin	• 30 –jährige Patientin • Granulomatose mit Polyangiitis • Immunsuppression mit Cylophosphamid
Aktuelle Beschwerden	• Otitis media
Antibiotika	• Ampicillin/Sulbactam

Verlauf	• Verschlechterung – Mastoiditis mit Mastoidektomie • V.a. Meningitis
Mikrobiologie	• Blutkulturen: negativ • Liquor: ausstehend

Case- Welches Antibiotikum empfehlen Sie?



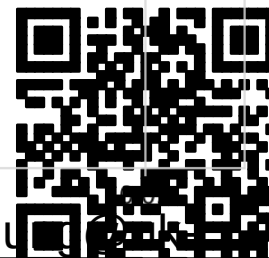
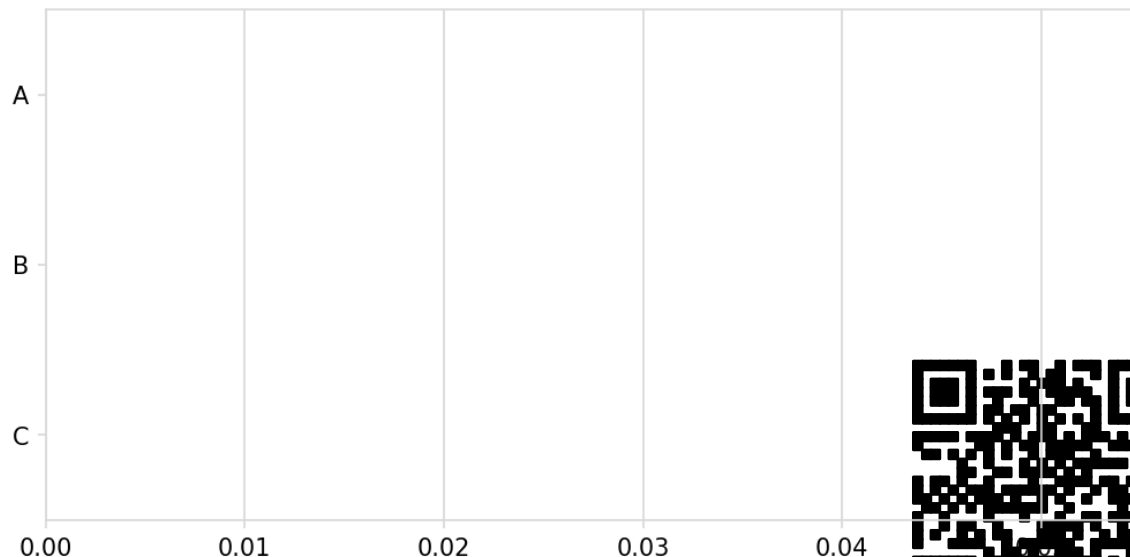
Madlener.. Jung Infection 2023



A: Ampicillin,
Ceftriaxon

B: Vancomycin,
Ceftriaxon

C: ein anderes



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Umfrage noch nicht



Case

Behandlung	Ampicillin + Ceftriaxon
Mikrobiologie	Liquor: neg. Kultur, multiplex PCR (:<i>H. influenzae</i>, <i>N. meningitidis</i>, <i>S. agalactiae</i>, <i>S. pneumoniae</i>, <i>L. monocytogene</i>, <i>E. coli</i>) negativ
Verlauf	Weitere Verschlechterung

Ambulant erworbene bakterielle Meningoenzephalitis im Erwachsenenalter

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Tabelle 3: Initiale Antibiotikatherapie der bakteriellen Meningitis (ohne Erregernachweis) bei Erwachsenen

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ambulant erworben („community-acquired“), keine Immunsuppression	Cephalosporin Gruppe 3a ¹ plus Ampicillin ²
nach neurochirurgischer Operation oder Schädel-Hirn-Trauma	Vancomycin plus Meropenem <i>oder</i> Vancomycin plus Ceftazidim ³ (plus Metronidazol bei operativem Zugang durch Schleimhäute)
immunsupprimierte Patientinnen und Patienten	Vancomycin plus Meropenem ⁴
Shunt-Infektion	Vancomycin plus Meropenem <i>oder</i> Vancomycin plus Ceftazidim

- Ambulant erworben

- Nach Op/Trauma

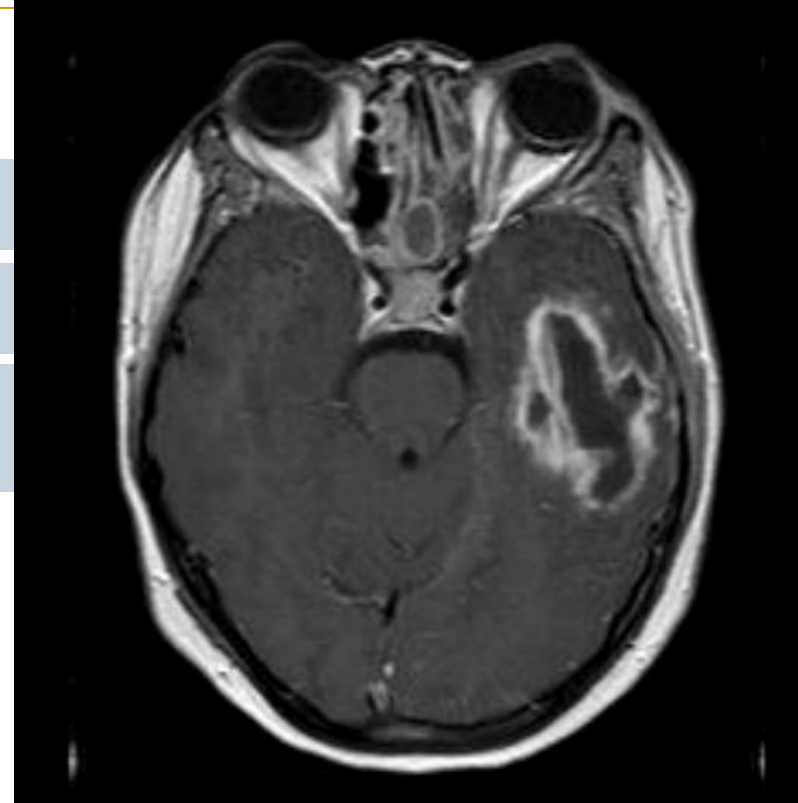
- Immunsupprimiert

- Shunt- assoziiert

Case

Behandlung	Meropenem + Vancomycin
Mikrobiologie	Liquor/Blutkulturen negativ
Verlauf	Weitere Verschlechterung

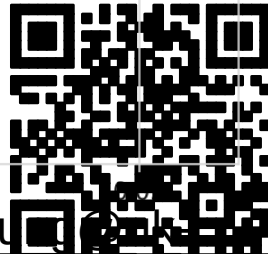
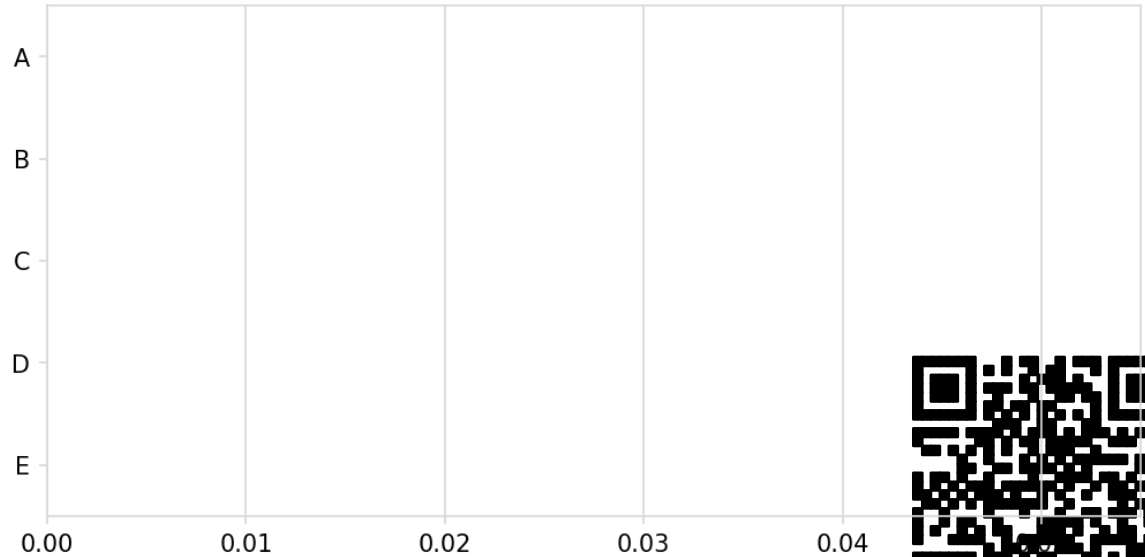
cMRI:
liquide Formation
52 x 25 mm



Case- Welchen Erreger vermuten Sie?



- A: Mycobacteria
- B: Fungi
- C: *S. aureus*
- D: *Toxoplasma* spp
- E: andere



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Umfrage noch nicht

Brain abscess with *Ureaplasma parvum* in a patient with granulomatosis with polyangiitis

Marie Madlener¹, Marianne Breuninger², Arne Meißner³, Henning Stetefeld⁴,
Sergej Telentschak⁵, Thorsten Wille⁶, Thilo van Eimeren⁴, Norma Jung²

Mikrobiologie	Hirnabszess: 2x PCR: <i>ureaplasma parvum</i>
Behandlung	Doxycyclin + Moxifloxacin
Verlauf	Heilung



Table 8. Duration of antimicrobial therapy for bacterial meningitis based on isolated pathogen (A-III).

Microorganism	Duration of therapy, days
<i>Neisseria meningitidis</i>	7
<i>Haemophilus influenzae</i>	7
<i>Streptococcus pneumoniae</i>	10–14
<i>Streptococcus agalactiae</i>	14–21
Aerobic gram-negative bacilli ^a	21
<i>Listeria monocytogenes</i>	≥21

Shorter is better



<https://www.bradspellberg.com/shorter-is-better>

Sudo, European J pediatrics, 2024

Shorter Is Better

Diagnosis	Short (d)	Long (d)	Result	#RCT
CAP	3-5	5-14	Equal	14
Atypical CAP	1	3	Equal	1
Possible PNA in ICU	3	14-21	Equal	1*
VAP	5 d	10-15	Equal	2

Bacterial Meningitis (peds)

4-7

7-14

Equal

6

cUTI/Pyelonephritis	5 or 7	10 or 14	Equal	11**
Intra-abd Infection	4	8-10	Equal	3
Complex Appendicitis	1-2	5-6	Equal	2
Bacteremia (non <i>S. aureus</i>)	7	14	Equal	4†
Cellulitis/Wound/Abscess	5-6	10	Equal	4‡
Osteomyelitis	42	84	Equal	2
Osteo Removed Implant	28	42	Equal	1
Debrided Diabetic Osteo	10-21	42-90	Equal	2 [¶]
Septic Arthritis	14	28	Equal	1
Bacterial Meningitis (peds)	4-7	7-14	Equal	6
AECB & Sinusitis	<5	>7	Equal	>25
Variceal Bleeding	2-3	5-7	Equal	2
Neutropenic Fever	AFx72h/3 d	+ANC>500/9 d	Equal	2
Post Op Prophylaxis	0-1	1-5	Equal	57 [¶]
Erythema Migrans (Lyme)	7-10	14-20	Equal	3
<i>P. vivax</i> Malaria	7	14	Equal	1

Choosing wisely – origin

Choosing wisely

- 2012, USA
- Verschiedene m
- Zu viel Behandlu

Klug entscheide

- 2014, Deutschland
- Deutsche Gesells
- (DGIM)
- Zu viel/zu wenig Behandlung

Schwerpunkt: Klug entscheiden

Internist
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„Klug entscheiden“ bei Infektionskrankheiten

Zu häufig Antibiotika – zu wenig Impfungen



Positiv-Empfehlungen

Choosing Wisely®

An initiative of the ABIM Foundation



Klug entscheiden

Eine Initiative der Deutschen Gesellschaft für Innere Medizin e.V. (DGIM)



Infection

<https://doi.org/10.1007/s15010-023-02005-y>

REVIEW

Choosing Wisely internationally – helpful recommendations for antimicrobial stewardship!

Norma Jung¹ · Lukas Tometten¹ · Rika Draenert²

213 Empfehlungen

Diagnostik (N=78; 40%)

Indikation (N=85; 37%)



Choosing wisely – Meningitis – nicht immer CT



Negativ- Empfehlungen

Eine **CT**-Untersuchung **vor der Lumbalpunktion** bei Verdacht auf bakterielle Meningitis soll **nicht** erfolgen – **außer** bei Symptomen, die auf einen erhöhten **Hirndruck** oder eine **Herdpathologie** hinweisen, oder bei **intensiver Immunsuppression**.

Grade A It is strongly recommended to perform cranial imaging before lumbar puncture in patients with:

- Focal neurologic deficits (excluding cranial nerve palsies).
- New-onset seizures.
- Severely altered mental status (Glasgow Coma Scale score <10).
- Severely immunocompromised state.

In patients lacking these characteristics, cranial imaging before lumbar puncture is not recommended.

Choosing wisely – Schneller Start



Positiv- Empfehlungen

Erwachsenen Patienten mit Verdacht auf bakterielle Meningitis sollen **nach der Blutkulturentnahme** noch **vor** einer eventuellen **Bildgebungsdiagnostik Dexamethason und Antibiotika** verabreicht werden.

Grade A

It is strongly recommended to start antibiotic therapy as soon as possible in acute bacterial meningitis patients. The time period until antibiotics are administered should not exceed 1 hour. Whenever lumbar puncture is delayed, e.g. due to cranial CT, empiric treatment must be started immediately on clinical suspicion, even if the diagnosis has not been established.

Zusammenfassung

- Empirische Therapie – nach Risiko und Resistenzlage
- Keine einheitliche Schwelle – je schwerwiegender desto niedriger anzusetzen
- Länge für verschiedene Erreger unterschiedlich
- An seltene Erreger denken
- Choosing wisely initiative– “Klug entscheiden”
 - Meningitis: sofort therapieren, nicht immer CT

